Guruprasath Gopal

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Education

National University of Singapore

August 2026 - August 2027

Master of Technology in Software Engineering | NUS-ISS

Singapore University of Technology and Design

September 2021 - May 2025

Bachelor of Engineering (Computer Science & Design) | Honours with Merit

Relevant Experience

Software Engineer | Data & Observability | BugRaid AI (Incident Management Startup)

August 2025 - June 2026

- Designed and built the core data processing pipeline for scalable ingestion and transformation of MELT data (Metrics, Events, Logs, Traces) from SigNoz, AWS S3, Loki, Prometheus, and custom sources.
- Containerised and deployed application services to AWS ECS through Amazon ECR, supporting repeatable and production-ready releases.
- Configured AWS autoscaling and cloud infrastructure to improve service availability and handle changing workloads reliably.

Software Engineer Intern | Primech A&P Pte Ltd

April 2024 - September 2024

- Built UX improvements for a PHP-based personnel management app that increased user update frequency by 15%
- Documented legacy codebase (API references, architecture, inline comments) to improve maintainability.
- Set up GitHub Actions workflows with Runbooks, reducing deployment time from 20 minutes to 3 minutes.

Robotics Engineer Intern (R&D Dept) | Primech A&P Pte Ltd

August 2023 - January 2024

- Enhanced robot dashboard with RESTful APIs and JWT authentication; Added real-time data feeds, performance metrics, and operational status displays to the dashboard, significantly improving user experience.
- Developed a Python backend for an IoT platform, ensuring seamless sensor interaction for IOT sensors.

Teaching Assistant | (Intro to Python Programming) | SUTD

November 2021 - Jan 2022

- Taught Python fundamentals to 40+ students; Set up Raspberry Pi IoT labs and resolved software/hardware issues
- Guided students through game development projects and coursework, delivering feedback and supplementary materials.

Projects

ResearchAgent - Budget-Aware Research Agent

- Built a payment-investigation agent combining deterministic transaction analysis, BM25 policy retrieval, function calling, grounded citations, and a no-LLM fallback.
- Built an agent that identifies which premium research evidence is worth buying within a user-approved source allowlist and budget.

TalkTolt - Local-First Health Analytics

- Built a parallel Apple Health ingestion pipeline using memory-mapped XML scanning, Parquet shards and DuckDB reconciliation, with coverage checks and staged database activation.
- Developed a local-first health analytics app with FastAPI, React and TypeScript, combining natural-language queries, structured dashboards, workout history and route exploration.

Plotline - Compact Geospatial Decision Layers

- Built a compact geospatial decision layer using bit-plane encoding and sparse events to preserve plot-screening outputs.
- Optimised polygon queries and registered-plot rechecks using cell indexing, compact decision codes and histogram aggregation.

Certifications

- AWS SOLUTIONS ARCHITECT - PROFESSIONAL SAP-C02 - June 2026
- AWS SOLUTIONS ARCHITECT - ASSOCIATE SAA-C03 - April 2026
- AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 - April 2026

Technical Skills

- Languages: Python, TypeScript, JavaScript, C++, Go, SQL, NoSQL
- Frameworks/Tools: React, Node.js, Express, GraphQL, Docker, Git, CI/CD
- LLM & AI: LangGraph, AWS Bedrock, BM25 retrieval, tool/function calling, prompt engineering, LLM evaluation, Langfuse
- Cloud & Quality: AWS (S3, CloudWatch , Lambda , SQS, Kinesis , AgentCore) , Jest, REST APIs , System Design.

Achievement & Awards

3x Hackathon Winner | 2025-2026

- InspireCon 2025 - 1st Place, DesignAI Award (Fitback). Award presented by President Tharman Shanmugaratnam; featured in The Business Times.
- SingHacks 2026 - Hackathon Winner (ResearchAgent); Ripple/XRPL Track Finalist and Singapore FinTech Festival Finalist.
- Common Ground Makeathon 2026 - Best Use of Tech (solo entry); awarded $1K in Codex credits.